

Jisung Yoo

Senior Research Scientist | 3D Computer Vision

South Korea | jisung.lucas.yoo@gmail.com

linkedin.com/in/jisung-yoo-7b340662 | jisung3d.github.io

PROFILE

3D vision researcher at NAVER LABS specializing in large-scale reconstruction, neural rendering, and generative 3D. Contributions span NAVER Map Street View 3D, Flying View 3D, city-scale digital twins, and media/VFX. Current research aims to enable visual simulation for robotics and autonomous driving through scene extension beyond captured trajectories and geometry-preserving appearance editing.

PROFESSIONAL EXPERIENCE

NAVER LABS

Feb 2022 - Present

Senior Research Scientist | Seongnam, South Korea

- **NAVER Map Street View 3D:** multi-view stereo depth reconstruction for multi-camera mapping rigs, RGB-D panoramic stitching, and mesh generation and simplification.
- **NAVER Map Flying View 3D:** dense matching and MVS preprocessing for aerial imagery, geometry-aware 3DGS reconstruction, city-scale tiled 3DGS training, and seamless LoD rendering.
- **Media & VFX:** dense 3D reconstruction and 3DGS training and rendering for real-world urban scenes.
- **City-scale digital twins:** MVS-based dense point-cloud and mesh reconstruction, with 3DGS-based scene reconstruction.
- **Current research:** extending scenes beyond captured trajectories for visual simulation in robotics and autonomous driving using 3D Gaussian Splatting, generative video/3D models, and feed-forward 3D reconstruction, with an emphasis on geometric and multi-view consistency. Parallel work explores geometry-preserving appearance editing.

Selected public work: [Flying View 3D](#) | [Street View 3D](#) | [Media & VFX](#)

Patents: three Korean patent applications filed at NAVER LABS (2025-present), pending publication.

Samsung Research

Jan 2021 - Jan 2022

Creative Leader, Team ReMo (Reconstruct Moments) | Seoul, South Korea

- Led an incubation team exploring AR/VR applications and business models for 3D content creation from smartphone photos.
- Prototyped photorealistic 3D-content experiences, including single-image animatable avatars and interactive 3D photos from smartphone image sequences.

Samsung Electronics

Mar 2018 - Dec 2020

Staff Engineer, Mobile Communications Division | Suwon, South Korea

- Developed core time-of-flight (ToF) AR measurement technologies for QuickMeasure, commercialized on the Galaxy S10 5G.
- Developed photo-filter style transfer for My Filters on the Galaxy S20 series; corresponding author of an ECCV 2020 Spotlight paper.
- Implemented the on-device Photo Remaster pipeline for degradation detection, automatic enhancement, and result recommendation, commercialized on the Galaxy S21 series.

RESEARCH BACKGROUND

KAIST

Mar 2010 - Feb 2018

Doctoral Research, Electrical Engineering | Daejeon, South Korea

- Implemented visual SLAM and volumetric reconstruction for multi-view 3D face reconstruction, with preprocessing for eye blinks, specular regions, and keyframe selection.
- Built RGB-D and multi-view capture systems; combined visual hulls, depth refinement, and view-dependent texture mapping for 3D object reconstruction.

SELECTED PUBLICATIONS

Filter Style Transfer between Photos

J. Yim, J. Yoo*, W.-J. Do, B. Kim, and J. Choe.

*European Conference on Computer Vision (ECCV), 2020. Spotlight. *Corresponding author.*

Visual Hull-Based Geometric Data Compression of a 3-D Object

Sung Soo Hwang, Wook-Joong Kim, Jisung Yoo, and Seong Dae Kim.

IEEE Transactions on Circuits and Systems for Video Technology, 25(7):1151-1160, July 2015.

Scale-invariant template matching using histogram of dominant gradients

J. Yoo, S. S. Hwang, S. D. Kim, M. S. Ki, and J. Cha.

Pattern Recognition, 47(9):3006-3018, September 2014.

TECHNICAL EXPERTISE

Research: 3D reconstruction, multi-view stereo, RGB-D reconstruction, visual SLAM, neural rendering, novel-view synthesis, 3D Gaussian Splatting, generative video/3D models, feed-forward 3D reconstruction.

Vision & imaging: Image processing, depth estimation, segmentation, image enhancement, style transfer.

Programming & platforms: C/C++, CUDA, OpenGL, Three.js, Android.

EDUCATION

KAIST

Mar 2010 - Feb 2018

Ph.D. in Electrical Engineering | Daejeon, South Korea

Dissertation: A study on interactive 3D reconstruction using RGB-D camera

KAIST

Mar 2006 - Jan 2010

B.Eng. in Electrical Engineering | Daejeon, South Korea

SELECTED RECOGNITION

Samsung Ph.D. Scholarship | Samsung Electronics | Mar 2013 - Feb 2017

Silver Prize, Korean Physics Olympiad (KPhO) | 2005